Clinical Practice Guidelines

Neutropenic Sepsis

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy

Introduction

This guideline provides direction to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the management of Neutropenic Sepsis.

Definitions

Fever	A range of differing opinions exist in relation to the definition of this term, but for the context of this document we will use 'single temperature equal to or greater than 38.3ºC or a sustained temperature equal to or greater than 38ºC for over a one-hour period'
Neutropenia	'neutrophil count less than 0.5 x 10^9 cells/L or less than 1.0 x 10^9 cells/L and predicted to fall below 0.5 x 10^9 cells/L'

1.0 Background

Neutropenic sepsis (or febrile neutropenia) is defined as:

- a single temperature equal to or greater than 38.3ºC; **or**
- a sustained temperature equal to or greater than 38ºC for over a one-hour period in a neutropenic patient (absolute neutrophil count less than 0.5 x 10^9 cells/L or less than 1.0 x 10^9 cells/L and predicted to fall below 0.5 x 10^9 cells/L).

Rapid initiation of broad-spectrum antimicrobials is essential for patients with neutropenic sepsis given their increased risk for acquiring severe infections and potential for clinical deterioration. Delays in antimicrobial administration or management may increase risk for morbidity and mortality.

2.0 Key Statements

Neutropenic sepsis (or febrile neutropenia), with or without fever is a MEDICAL EMERGENCY and should NOT be classified lower than an ED Triage Category 2.

- Management of neutropenic sepsis must be commenced urgently.
- Patients may present without fever (particularly if elderly or on steroids), but with other signs of sepsis.
- The administration of empiric antibiotics should ideally commence within 30 minutes of presentation (and should not be delayed to perform investigations) and within 60 minutes if clinically stable.

3.0 Neutropenic Sepsis Management Algorithm**

**Please refer to the Neutropenic Sepsis Guideline (4.0 onwards) for more comprehensive details regarding the management of this serious condition. If additional information or clarification is required contact a Clinical Haematologist or Infectious Diseases Physician for advice

NEUTROPENIC SEPSIS IS A MEDICAL EMERGENCY

INITIAL ASSESSMENT

- ☐ Patient History: source of infection, prior infection, antimicrobial prophylaxis, drug allergies, organ Impairment
- ☐ Patient examination: pulse, BP, RR, O2 saturation, chest, skin/mucous membranes, peri-anal

Do NOT perform a PR examination or administer medications via PR route

INITIAL INVESTIGATIONS

- ☐ Bloods: full blood picture, coagulation profile, urea/electrolytes, liver function tests, CRP, blood glucose
- ☐ Microbiology: blood cultures x 2 (central venous access device + peripheral), MSU, sputum culture (if productive), dry throat swab (if respiratory symptoms present) for viral PCR, faeces m/c/s (if diarrhoea), wound swabs (if applicable)
- ☐ Radiology/Imaging: chest X-ray and any additional imaging as indicated

STAGE 1 - INITIAL ANTI-INFECTIVE THERAPY

Preferably start antimicrobials after blood cultures taken however therapy should NOT be delayed unnecessarily. Doses provided are for normal renal function. *Check eTG for dosage modifications if renal impairment exists.*

piperacillin-tazobactam* 4.5g IV 6 hourly

***If delayed penicillin hypersensitivity – use cefepime 2g IV 8 hourly (Add metronidazole 500mg IV 12 hourly if suspicion of intra-abdominal infection)**
***If severe penicillin hypersensitivity eg anaphylaxis, Steven Johnsons Syndrome/Toxic Epidermal Necrolysis (SJS/TENS) – use vancomycin IV (refer to SCGH vancomycin guideline) PLUS ciprofloxacin 400mg IV 12 hourly. Add metronidazole IV if suspicion of intra-abdominal infection**
+If concerns for nephrotoxicity relating to piperacillin-tazobactam/vancomycin combination therapy substitute piperacillin-tazobactam with cefepime 2g IV 8 hourly (add metronidazole IV if suspicion of intra-abdominal infection)

SYSTEMIC COMPROMISE?

(e.g. Systolic Blood Pressure (SBP) < 90mmHg, vasopressor support required, major organ dysfunction etc)

YES

- Continue piperacillin-tazobactam.
- Consider addition of vancomycin+ IV PLUS gentamicin IV STAT dose (4-7mg/kg daily as per eTG).

NO

- Continue piperacillin-tazobactam therapy

OTHER IMPORTANT CONSIDERATIONS

Suspected or known gram-positive infection?

If YES
Add vancomycin+ (if not already initiated). Review ongoing use after 48 to 72 hours

Suspected multi-resistant gram-negative infection?

If YES
Contact a Clinical Microbiologist or ID Physician for advice

Suspicion of pneumonia?

If YES
Add azithromycin 500mg IV daily for 3 days. Consider influenza, COVID-19 and PJP testing and treatment

Suspicion of *C. difficile* colitis?

If YES
Send faeces for *C. difficile* testing. Treat as per eTG recommendations

Neutropenic Sepsis Management Algorithm

HAPi
Healthy Antimicrobial Prescribing Initiative

STAGE 2 – FOLLOWING INITIAL 24-48 HOURS OF ANTI-INFECTIVE THERAPY

If patient deteriorates clinically (e.g. haemodynamic instability, O_2 desaturation) whilst receiving empiric antimicrobial regimen

- Consider any gaps in current antimicrobial regimen (e.g. gram-positive, gram-negative, anaerobes)
- Contact the Infectious Diseases team for advice

If patient has ongoing fevers but is clinically stable with NO focus of infection and NO concerns regarding piperacillin-tazobactam resistance

- Continue current therapy WITHOUT escalating antimicrobials

If a plausible pathogen or source of infection is identified

STAGE 3

IF CLINICAL DETERIORATION OR PERSISTENT FEVER AFTER 96 HOURS OF BROAD SPECTRUM ANTIBIOTIC COVER AND EXPECTED DURATION OF NEUTROPENIA >7 DAYS

Consider invasive fungal disease

Perform additional tests and consider treatment options in consultation with ID#

If signs of invasive mould infection

If receiving fluconazole or nil fungal prophylaxis

Suggest either
1. Liposomal amphotericin^ **3mg/kg** IV daily.
OR
2. Voriconazole^† IV 6mg/kg 12 hourly for first 24 hours then 4mg/kg 12 hourly thereafter.

If receiving voriconazole/ posaconazole prophylaxis

Suggest
1. Liposomal amphotericin^ **5mg/kg** IV daily.

If NO signs of invasive fungal infection

Consider empirical antifungal treatment for any patient considered high risk for fungal infection (severe oral/GI mucosal injury, CVAD in situ).
Choice of antifungal agent will be guided by clinical features and ID input.

^ Requires ID/Micro approval prior to starting

† Preferred option if high risk for Aspergillosis

STAGE 4 – IF CLINICALLY IMPROVING

De-escalation of therapy

Consideration of de-escalation should be discussed with the Treating Specialist before making any changes. Consider if:

- ✓ Patient is clinically and haemodynamically stable
- ✓ Afebrile for 72 hours
- ✓ Able to swallow and absorb oral medications
- ✓ Evidence of neutrophils recovering despite count still <1.0

Suggest

- Oral amoxicillin-clavulanate 875mg/125mg twice a day,

PLUS

- Oral ciprofloxacin 500mg twice a day.

Note: This suggestion is an empiric option only. If a plausible pathogen or source identified, then treatment may require amendment. Dosage modifications may be required depending on other factors (pathogen, organ dysfunction etc)

4.0 Initial Assessment

4.1 Patient history

- Potential source of current infection.
- Prior infections (e.g. Methicillin-Resistant Staphylococcus aureus (MRSA), other multi-resistant organisms, fungal).
- Pre-existing antimicrobial prophylaxis.
- Drug allergies.
- Renal/hepatic impairment.

4.2 Examination

- Pulse.
- Blood Pressure.
- Respiratory Rate.
- Oxygen saturation.
- Chest sounds.
- Central venous access device (CVAD) site.
- Skin and mucous membranes.
- Peri-anal**

****do NOT perform a PR examination or administer medications via PR route due to risk of provoking a bacteraemia from a rectal source**

5.0 Initial Investigations

5.1 Bloods

- Full blood picture.
- Coagulation profile.
- Urea and electrolytes.
- Liver Function Tests (LFTs)
- Blood glucose level.
- CRP.

5.2 Microbiology

- Blood cultures (minimum of 2 sets with 8-10 mL of blood added to each bottle (aerobic and anaerobic) collected from CVAD and peripheral sites. Refer to the SCGOPHCG Blood cultures: when and how to take guideline for more information.
- MSU (Mid-Stream Urine).
- If producing sputum request for bacteria, fungal, nocardia, mycobacteria – microscopy and culture PLUS viral, atypical bacteria and Pneumocystis jiroveci pneumonia (PJP) PCR.
- If not producing sputum then perform a dry throat swab and request viral and atypical bacteria PCR.
- If diarrhoeal disease: faeces m/c/s, *C. difficile* and viral PCR.
- Other cultures from any other suspected source of infection.
- If clinical or epidemiological risk for COVID-19 then perform testing as per current COVID-19 guidelines:
- SCGOPHCG Specimen Collection for Microbial Examination Guideline: available here.

5.3 Radiology

- Chest X-ray and additional imaging as clinical indicated.

6.0 Anti-infective Management

6.1 Empiric treatment approach

Preferably start antibiotics after blood cultures have been collected, however do not delay antibiotic administration unnecessarily whilst pending blood culture collection. Patients with signs of sepsis, or who present unwell and are assessed as potentially septic by a senior clinician, should receive antibiotics within 30 minutes of arrival.
Doses provided are for normal renal function. Check eTG (Electronic Therapeutic Guidelines) if renal impairment present.

INITIAL THERAPY
piperacillin-tazobactam* 4.5g IV 6 hourly

*If delayed penicillin hypersensitivity, use cefepime 2g IV 8 hourly. Add in metronidazole 500mg IV 12 hourly if there is suspicion of intra-abdominal source of infection.

*If immediate penicillin hypersensitivity (eg urticarial, angioedema, bronchospasm or anaphylaxis), use vancomycin IV (refer to SCGOPHCG vancomycin guideline) with ciprofloxacin 400mg IV 12 hourly. Additional metronidazole 500mg IV 12 hourly if there is suspicion of intra-abdominal source of infection.

+If concerns for nephrotoxicity relating to piperacillin-tazobactam/vancomycin combination therapy (e.g. pre-existing renal failure and additional nephrotoxic agents), consider substituting piperacillin-tazobactam with cefepime 2g IV 8 hourly (+/- metronidazole IV) or consulting with ID. Patients on cefepime should be monitored closely for encephalopathy, particularly if there is with pre-existing renal failure.

6.2 IMPORTANT CONSIDERATIONS

6.2.1 **If systemic compromise (e.g. Systolic Blood Pressure (SBP) < 90mmHg, requiring vasopressor support, confusion, DIC, new or worsening major organ dysfunction, O_2 saturation < 90%)**

- Consider addition of vancomycin+ IV and gentamicin# IV.
- # Gentamicin 4-7mg/kg IV daily (depending on renal function) – Refer to the eTG (Figure 2.103) for contraindications and precautions relating to the use of aminoglycosides.

6.2.2 **If suspected or known gram-positive infection (e.g. gram positive organism in blood culture, MRSA / VRE (Vancomycin-Resistant Enterococci) colonisation, skin or soft tissue source of infection, presence of suspected infected intravascular device in situ)**

- Add vancomycin IV+.
- Empiric vancomycin IV therapy should be reviewed after 48 to 72 hours. If culture and susceptibility results return and there is no longer any indication to continue vancomycin, therapy should be ceased provided that the patient is clinically improving.

6.2.3 **If suspected multi-resistant gram-negative infection**

- e.g. ESBL (Extended Spectrum Beta-Lactamase) or carbapenem-resistant *Enterobacteriaceae* colonisation, resistant gram-negative organism previously isolated, current/recent antimicrobial therapy or Pseudomonal culture.
- Contact a Clinical Microbiologist or Infectious Diseases Physician for advice.

6.2.4 **If pneumonia is present**

- Add azithromycin 500mg IV daily.

- Consider influenza, SARS-CoV-2 (COVID-19) and PJP testing and treatment.
- Consider Respiratory Team consultation (including opinion on suitability for bronchoscopy).
- If viral respiratory infection is suspected, droplet precautions must be commenced (refer to Transmission Based Precautions Policy).

6.2.5 **If *C. difficile* colitis suspected**

- Send faeces specimen for *C. difficile* testing.
- Treat as per eTG recommendations.

6.2.6 **If colonisation with a resistant pathogen, recent/current antimicrobial therapy, neutropenia period > 7 days, previous invasive fungal infection or likely focal site of infection:**

- Contact a Clinical Microbiologist or Infectious Diseases Physician for advice.

7.0 Modifications to the Antimicrobial Regimen (FOLLOWING INITIAL 24-48 HOURS OF ANTI-INFECTIVE THERAPY)

7.1 If the patient deteriorates clinically (e.g. haemodynamic instability, O_2 desaturation) whilst receiving the empiric antimicrobial regimen

- Consider any gaps in the current antimicrobial regimen the patent is receiving (e.g. gram-positive, gram-negative, anaerobes, atypical).
- Contact the Infectious Diseases team to discuss therapeutic options and for advice.
- If restricted therapy is deemed necessary, ID/Micro-approval will be required before commencement.

7.2 If patient has ongoing fevers but remains clinically stable with no focus of infection and no concerns for piperacillin-tazobactam/cefepime resistance

- Continue current therapy without escalating antimicrobials.
- Avoid escalating antimicrobial therapy if the patient is haemodynamically stable, despite the presence of any fever.

7.3 If a plausible pathogen or source of infection is identified

- Consider targeted antimicrobial therapy and treat for the recommended duration for the specific infection type.
- NB do not narrow antimicrobial spectrum if concurrent sources of infection present.

8.0 Modifications to the Antimicrobial Regimen (FOLLOWING 96 HOURS OF ANTI-INFECTIVE THERAPY)

8.1 Consider invasive fungal infection (IFI) if

- There is clinical deterioration or persistent fever despite 96 hours of broad spectrum antibiotic cover, particularly in high risk^ patients.
  ^ haematological malignancy, expected duration of neutropenia exceeding 7 days.

8.2 Additional suggestions

- Perform CT chest and sinuses.
- Serum galactomannan.
- Blood cultures (to investigate for candidaemia).
- Continue to investigate for other sources of fever (in consultation with Infectious Diseases or Clinical Microbiology teams):
  - Respiratory tract specimens (including bronchoscopy or lung biopsy if sputum samples are negative).
  - Tissue biopsy of other relevant sites.
  - CT imaging (e.g. abdomen).
  - Faeces specimen (e.g. *C. difficile* toxin gene PCR).

  - Plasma CMV viral load.
  - Other causes: thrombophlebitis, haematoma, drug-related fever, malignancy-related fever.

9.0 Empiric Anti-fungal Treatment Approach

9.1 If signs of invasive mould infection whilst receiving either fluconazole or nil fungal prophylaxis therapy

Suggestions:

- Liposomal amphotericin 3mg/kg IV daily,
  OR
- Voriconazole IV (6mg/kg 12 hourly for first 24 hours, then 4mg/kg 12 hourly thereafter.

Voriconazole is the preferred option if the patient is considered high risk for Aspergillosis (consult ID for advice).

9.2 If signs of invasive mould infection whilst receiving voriconazole or posaconazole prophylaxis therapy

Suggestion:

- Liposomal amphotericin 5mg/kg IV daily.

9.3 If no signs of invasive mould infection, however still considered at high risk for fungal infection (eg severe oral/GI mucosal injury, CVC in situ)

- Consider empiric antifungal therapy.
- Choice of antifungal agent should be guided by clinical features together with ID advice.

10.0 De-escalation of Therapy

De-escalation of antimicrobial therapy should be considered following patient improvement or clinical progression. Any decision to de-escalate or adjust antimicrobial therapy should be discussed with the treating specialist/s prior to any changes.

De-escalation may be considered if:

- Patient is clinically and haemodynamically stable.
- Afebrile for ≥ 72 hours.
- Able to swallow and absorb oral medications.
- Evidence of neutrophil recovery despite neutrophil count still < 1.0

If considered appropriate for oral de-escalation therapy, suggest

- Oral amoxicillin-clavulanate 875mg/125mg twice a day,
  PLUS
- Oral ciprofloxacin 500mg twice a day.

This suggestion is an empiric option and should be modified appropriately if a plausible pathogen is detected. Dosage modifications may be required depending on other factors (pathogen, organ dysfunction etc).

Related Documents

Legislation:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016

- Health Services Act 2016

Policy and Standard/s:

- Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy
- SCGOPHCG Formulary Management Procedures MMP013
- SCGOPHCG Sepsis Management CPG
- SCGOPHCG High Risk Medications policy
- SCGOPHCG Medicines Management Policy
- SCGOPPHCG Medication Management Guideline: Vancomycin IV
- SCGOPHCG Standard and Transmission Based Precautions Policy
- SCGOPHCG COVID-19 policy
- SCGOPHCG Specimen collection guideline
- SCGOPHCG Blood cultures: when and how to take guideline
- eTG: electronic Therapeutic Guidelines: Antibiotic

References

1. Averbuch D, et al. European guidelines for empirical antibacterial therapy for febrile neutropenic patients in the era of growing resistance: summary of the 2011 4th European Conference on Infections in Leukemia. Haematologica 2013; 98 (12): 1826-1835.
2. Freifeld A, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the Infectious Diseases Society of America. Clin Infect Dis 2011;52(4):e56-e93.
3. Marr K. Treatment and prevention of invasive aspergillosis. UpToDate. Marr K. Diagnosis of invasive aspergillosis. UpToDate.
4. Slavin M, et al. Introduction to the updated Australian and New Zealand consensus guidelines for the use of antifungal agents in the haematology/oncology setting, 2008. Int Med J 2008;38:457-467.
5. Wingard JR. Treatment of neutropenic fever syndromes in adults with hematologic malignancies and hematopoietic cell transplant patients (high-risk patients). UpToDate.
6. S. Lingaratnam et al. Introduction to the Australian consensus guidelines for the management of neutropenic fever in adult cancer patients, 2010/2011. Internal Medical Journal 41(2011) 75-81.

Authorisation & Version Control

Policy Sponsor:	Medical Co-Director CICS				
Policy Contact:	Senior Infectious Diseases/Antimicrobial Stewardship Pharmacist				
Version #:	4	**Last Reviewed:**	09/09/2025	**Review Due:**	09/09/2026
Reference #	CPG001	**Consumer Engagement:**	N/A	**Aboriginal Health ISD #:**	N/A
Approved by:	Medical Co-Director CICS			**Date:**	21/10/2025
Endorsed by:	SCGOPHCG Clinical Practice Guideline Committee			**Date:**	09/09/2025
Applicable Standards	NSQHS Standards Std 1: Clinical Governance | Std 3: Preventing and Controlling Healthcare Associated Infection | Std 4: Medication Safety | Std 5: Comprehensive Care | Std 6: Communicating for Safety				
Printed or personally saved electronic copies of this document are considered uncontrolled					